Science Homework: Life Processes: nutrition and respiration

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Q1. What are the raw materials and the products of photosynthesis? Why does a plant need sunlight for it?

Photosynthesis is the process by which green plants synthesise carbohydrates. The raw materials are carbon dioxide, which diffuses in through the stomata, and water, which is absorbed by the roots. Chlorophyll captures light energy and converts it into chemical energy, enabling the reduction of carbon dioxide to glucose. Oxygen is liberated as a by-product. Thus sunlight provides the energy that drives the entire process.

Q2. Distinguish between aerobic and anaerobic respiration. Give one example of each.

Aerobic respiration occurs in the presence of oxygen; glucose is completely oxidised to carbon dioxide and water, yielding a large amount of energy (ATP). Example: cellular respiration in the mitochondria of human cells. Anaerobic respiration occurs in the absence of oxygen; glucose is only partially broken down, releasing less energy. Example: fermentation by yeast, which produces ethanol and carbon dioxide. Furthermore, muscle cells produce lactic acid during strenuous activity.

Q3. Why are the alveoli in our lungs richly supplied with blood capillaries?

The alveoli provide an enormous, thin surface area for gaseous exchange. A dense capillary network ensures that oxygen diffuses rapidly into the blood while carbon dioxide diffuses out, maintaining a steep concentration gradient and maximising the efficiency of the exchange of gases.